



SAINT LOUIS UNIVERSITY
MADRID

MATH 1660 M01: Discrete Mathematics
Spring 2025

Credit(s):	3
Prerequisite(s):	C- or better in MATH-1200 College Algebra (or equivalent)
In-Person Class Time:	MW, 14:00-15:15 (Madrid time)
Classroom:	PRH-A008
Instructor:	Luis Miguel Anguas
Instructor's Email:	luismiguel.anguas@slu.edu
Instructor's Phone:	(+34) 91 554 58 58, ext. 299
Office:	PAH-S01
Office Hours:	MW, 13:00-13:50, TR 11:15-12:25 (Madrid time) and by appointment

Course Description: Concepts of discrete mathematics used in computer science; sets, sequences, strings, symbolic logic, proofs, mathematical induction, sums and products, number systems, algorithms, complexity, graph theory, finite state machines.

Course Goals and Student Learning Outcomes: At the end of the course, students will:

- Reformulate statements from common language to formal logic and analyze their truth values.
- Give proofs of simple statements using methods such as direct proof or proof by contradiction.
- Use set theory to solve problems.
- Provide proofs using mathematical induction.
- Work with graphs that includes translating simple problems into graph theoretic ones.

Technology Requirements: You will need regular access to a computer with an internet connection. High speed broadband access (LAN, Cable or DSL) is highly recommended for the optimal learning experience. Students should preferably use a computer, NOT a telephone or tablet to complete assignments and work.

Communication Norms: Preferred means of communication:

- For general questions that you would normally ask during an in-person class, use the "Q & A forum" of the Discussion Board of Canvas.
- For specific questions about navigating Canvas or for personal concerns, questions, or requests email me.
- Canvas will be used regularly for posting material, lectures, homework, making announcements, etc.
- You should check your "@slu.edu" email daily. Campus and course announcements will be often communicated via email.

Email reply times:

- I will aim to reply to your emails within 24 hours Monday through Friday. If you do not hear back from me after 24 hours, please resend the email so that it will be bumped up to the top of my inbox.
- I cannot guarantee that emails sent after 6:00 p.m., Madrid time, will receive replies until the following workday morning.
- Students are expected to respond to my emails within the same time frame.

Grades and Feedback:

- All tasks must be completed before their corresponding deadline and submitted according to the instructions of that task.

Netiquette: Certain behavioral codes are expected when you communicate with both your peers and your instructors in any learning environment. For online-course discussions and emails, these codes are referred to as netiquette. For further information, check SLU netiquette guidelines.

Assessment of Student Learning: To maintain quality academic offerings and to conform to accreditation requirements, SLU-Madrid regularly assesses its teaching, services and programs for evidence of student learning. For this purpose, SLU-Madrid keeps representative examples of student work from all courses and programs on file, including assignments, papers, exams, portfolios and results from student surveys, focus groups and reflective exercises. Copies of your work for this course may be kept on file for institutional research, assessment and accreditation purposes. If you prefer SLU-Madrid not to retain your work for this purpose, you must communicate this decision in writing to your professor.

Required Texts and Readings: None.

Optional Texts and Readings: R. Johnsonbaugh, *Discrete Mathematics*, Eighth Edition.

Engagement & Attendance Policies: Attendance and participation are very important in the learning process; missing classes will have a negative impact on your final grade. If an absence is unavoidable, you will be responsible for any announcements, information, problems, or course changes that are made in all lectures. Students are expected to arrive on time to the lectures. Repeated lateness will not be tolerated. So, please come to class on time and do not leave early. Anything else is rude and disruptive.

It is the student's responsibility to find out what was covered in class and to secure notes from another student. Copies of my personal notes are not to be shared with students. During office hours, and after the student has read and worked on the notes on his own, I will gladly answer any specific question regarding the material missed.

Good attendance by itself, however, is no guarantee of passing the class. Students should estimate to dedicate from 5 to 9 hours of work per week during the semester to complete each module.

Other Information on Course Assignments & Activities: Homework will be assigned for every topic covered. Solutions will be provided, and students are expected to check their exercises and come to office hours with any difficulties encountered.

Grading Rationale/System: Your final grade is calculated as follows:

- Homework: 20%
- In-class exams (two): 45%
- Final exam: 35%

Note: If you get less than 35% in your final exam, your final grade will be either an F or a D; in this case your final grade will be an F if the mean of the course is less than 55% and a D if the mean of the course is more than 55%.

Grading System: A = 93 or higher; A- = 90-92; B+ = 87-89; B = 83-86; B- = 80-82; C+ = 75-79; C = 70-74; C- = 65-69; D = 55-64; F = 55 or less.

Academic Integrity: Academic integrity is the commitment to and demonstration of honest and moral behavior in an academic setting. Since the mission of the University is "the pursuit of truth for the greater glory of God and for the service of humanity," acts of integrity are essential to its very reason for existence. Thus, the University regards academic integrity as a matter of serious importance. Academic integrity is the foundation of the academic assessment process, which in turn sustains the ability of the University to certify to the outside world the skills and attainments of its graduates. Adhering to the standards of academic integrity allows all members of the University to contribute to a just and equitable learning environment that

cultivates moral character and self-respect. The full University-level Academic Integrity Policy can be found on the Provost's Office website at:

<https://www.slu.edu/provost/policies/academic-and-course/academic-integrity-policy.pdf>.

Your instructor will review these matters during the first weeks of the term. Please direct questions about any facet of academic integrity to the instructor, the chair of the department of your academic program or the Academic Dean of the Madrid Campus.

AI Policy: The University's goals include fostering students' critical thinking, creativity, reflection and problem solving. Therefore, all students must demonstrate originality in the authorship of their work and properly cite the sources consulted. While proper use of AI may prove to be a valuable teaching tool for both teachers and students, improper use/abuse of AI might be considered plagiarism and penalized as such. Please consult with your instructor if you have questions about whether or not you can use AI tools such as ChatGPT in your course.

In order to develop forms of assessment that will allow students to be assessed in a reliable manner and to provide effective and enhanced support for their learning, Departments/instructors reserve the right to randomly conduct oral exams as a second stage of an assessment for which the first stage is the submission of a written assignment.

Diversity and Inclusion: Saint Louis University is committed to fostering a positive, inclusive and welcoming learning and working environment. SLU-Madrid's policies prohibit discrimination based on race, color, national origin, ancestry, religion, sex, gender, gender identity, gender expression, sexual orientation, pregnancy, age, disability, physical appearance, financial or socio-economic status, immigration status, parental or marital status, veteran status or any other protected classification of identity. Discrimination includes any form of unequal treatment such as denial of opportunities, harassment, and violence. Sex-based violence includes rape, sexual assault, unwanted touching, stalking, dating/interpersonal violence, and sexual exploitation.

If you experience or witness any kind of discrimination, you are encouraged (but not required) to report the incident to the SLU-Madrid's Department of Student Life (studentlife-madrid@slu.edu; +34 915 54 58 58, ext. 213) or, if you wish to speak to a confidential resource, the Counseling Center (counselingcenter-madrid@slu.edu +34 915 54 58 58, ext. 230). Both are in Padre Rubio Hall, on the ground floor and the first floor, respectively. You can also report the incident to the University's Hotline (900-99-0011; then enter 877-525-5669 when asked for the hotline number).

Please know that instructors are required to inform SLU-Madrid when made aware of incidents of discrimination, harassment sexual misconduct, and/or related retaliation, to ensure that individuals impacted receive information about options for reporting and support resources. If you wish to speak with a confidential resource, the following are available on campus and off campus:

- Counselors at the [Counseling Center](mailto:counselingcenter-madrid@slu.edu), located on the first floor of Padre Rubio Hall (counselingcenter-madrid@slu.edu; 915 54 58 58, ext. 230).
- Counselors at Sinews Multilingual Therapy Institute, SLU-Madrid's off-campus counseling and mental health services provider (www.sinews.es; 917 00 19 79).

If you prefer to seek confidential spiritual counseling related to an incident of discrimination or harassment, contact SLU-Madrid's Campus Chaplain, Fr. James O'Leary, S.J. (james.oleary@slu.edu; 915 54 58 58, ext. 279).

Additional information and resources are posted on our [Safety and Security](#) and [Community Standards](#) webpages.

Accessibility, Disability and Learning Resources: In recognition that people learn in a variety of ways and that learning is influenced by multiple factors (e.g., prior experience, study skills, learning disability), resources to

support student success are available on campus. Students who think they might benefit from these resources can find out more about:

- Course-level support (e.g., faculty member, departmental resources, etc.) by asking or emailing your course instructor.
- University-level support (e.g., tutoring/writing services, Accessibility and Disability Resources) by contacting the Academic Dean's Office (advising-madrid@slu.edu) or by reviewing the [Academic Resources website online](#).

Students with a documented disability who wish to request academic accommodations must contact Disability Services to discuss accommodation requests and eligibility requirements. Once successfully registered, the student also must notify the course instructor that they wish to access accommodations in the course. Please contact disabilityservices-madrid@slu.edu or +34 915 54 58 58, extension 242 or 249. Confidentiality will be observed in all inquiries. Once approved, information about the student's eligibility for academic accommodations will be shared with course instructors via email from Disability Services. Note: Students who do not have a documented disability but who think they may have one are encouraged to contact Disability Services.

For more information about academic accommodations, see [SLU-Madrid's Center for Accessibility and Disability Resources webpage](#).

Needs Security Statement: Students in personal or academic distress and/or who may be specifically experiencing challenges such as securing food or difficulty navigating campus resources, and who believe this may affect their performance in the course, are encouraged to contact Student Life (studentlife-madrid@slu.edu or +34 915 54 58 58, ext. 213) for support. Furthermore, please notify the instructor if you are comfortable in doing so, as this will enable them to assist you with finding the resources you may need.

Use of Posted Course Content: SLU-Madrid prohibits recording and transmission of classroom lectures and discussions by students unless written permission from the class instructor has been obtained and all students in the class as well as guest speakers have been informed that audio/video recording may occur. Recordings, course materials, and lecture notes may not be exchanged or distributed for commercial purposes, for compensation, or for any other purpose other than study by students enrolled in the class. Public distribution of such materials may constitute copyright infringement in violation of Spanish law. Violation of this policy may subject a student to disciplinary action in accordance with SLU-Madrid policies on [Student Rights and Responsibilities and Community Standards](#).

GDPR Norms Concerning Class Recordings: In accordance with General Data Protection Regulation (GDPR), we inform you that if you are a participant in an on-line classroom, your image and voice may be recorded by Saint Louis University in Spain, S.A., CIF A28654879, in 28003 Madrid (Spain), Avenida del Valle 34, for the sole purpose of the instruction of the said class that you are registered in. This information will be stored for the duration of the on-line class and erased thereafter by the professor of the course. Should you not want your image or voice to be a part of this class recording, please contact your professor to indicate that you will be turning your camera and microphone off and be participating via chat.

In addition, we would like to inform you that all recordings will be available to you in Canvas and are exclusively for the use of the participants of the said class and should not be published on any other platform without the prior consent of all participants that may appear in the recording.

According to the rights conferred by the current GDPR regulation, you may exercise your rights of access, rectification, limitation of treatment, deletion, portability and opposition to the processing of your personal data, as well as the consent given for the treatment of it by directing your requests to the address indicated above or by sending an email to dpo-madrid@slu.edu.

Spring 2025 Course Outline

1. Set theory.
2. Logic.
 - a. Propositions.
 - b. Conditional propositions.
 - c. Quantifiers.
 - d. Nested quantifiers.
3. Proofs.
 - a. Mathematical systems.
 - b. Methods of proofs.
 - i. Direct proofs.
 - ii. Counterexamples.
 - iii. Proofs by contradiction.
 - iv. Proof by cases.
 - v. Proofs of equivalence.
 - vi. Existence proofs.
 - vii. Mathematical Induction.
 - viii. Common errors in proofs.
4. Sequence and relations.
 - a. Sequences.
 - b. Relations.
 - c. Equivalence relations.
5. Counting arguments.
 - a. Permutations.
 - b. Variations.
 - c. Combinations.
 - d. Combinatorial formulas with repetitions.
6. Graph theory.
 - a. Paths and cycles.
 - b. Hamiltonian cycles.
 - c. A shortest-path algorithm.
 - d. Representation of graphs.
 - e. Isomorphisms of graphs
 - f. Planar graphs.

First midterm: February, 19th

Second midterm: March, 19st

Final exam: May, 7th (13:30-15:20)

Spring 2025 Important Dates:

Tuesday, Jan. 7	Move-in day new and study abroad students
Wednesday, Jan. 8	Welcome sessions
Thursday, Jan. 9	First day of classes
Sunday, Jan. 19	Last day to drop a class without a grade of W and/or add a class Last day to choose Audit (AU) or Pass/No Pass (P/NP) options
Wednesday, Feb. 19	Registration for summer sessions begins (4 p.m., Madrid time)
Thursday-Friday, Feb. 20-21	Winter break
Sunday, March 2	Professors' deadline to submit midterm grades
Sunday, March 16	Last day to drop a class and receive a grade of W
Sunday, April 13	<i>Semana Santa</i> host family housing move-out day
Monday-Sunday, April 14-20	<i>Semana Santa</i> (Madrid campus closed)
Sunday, April 20	<i>Semana Santa</i> host family housing move-in day
Tuesday, April 22	Priority registration for fall semester begins (4 p.m., Madrid time)
Wednesday, April 23	Registration for fall semester begins (4 p.m., Madrid time)
Wednesday, April 30	Last day of classes
Thursday, May 1	<i>Día del Trabajador</i> (Madrid campus closed)
Friday, May 2	<i>Día de la Comunidad de Madrid</i> (Madrid campus closed)
Monday-Friday, May 5-9	Final exams
Friday, May 9	Awards Ceremony Host family housing move-out day
	Commencement (TBA)
Monday, May 12	Professors' deadline to submit final grades

Final Exam Schedule Spring 2025

Class Meeting Time*	Exam Date/Time
Mondays 8 a.m.	Thursday, May 8, 8:30 to 10:20 a.m.
Mondays 9 a.m.	Tuesday, May 6, 8:30 to 10:20 a.m.
Mondays 10 a.m.	Thursday, May 8, 11 a.m. to 12:50 p.m.
Mondays 11 a.m.	Tuesday, May 6, 11 a.m. to 12:50 p.m.
Mondays noon	Thursday, May 8, 1:30 to 3:20 p.m.
Mondays 1 p.m.	Tuesday, May 6, 1:30 to 3:20 p.m.
Mondays 2 p.m.	Wednesday, May 7, 1:30 to 3:20 p.m.
Mondays 3:30 p.m.	Thursday, May 8, 4 to 5:50 p.m.
Mondays 5 p.m.	Tuesday, May 6, 4 to 5:50 p.m.
Mondays 6:30 p.m.	Thursday, May 8, 6:30 to 8:20 p.m.
Mondays 8 p.m.	Tuesday, May 6, 6:30 to 8:20 p.m.
Tuesdays 8 a.m.	Monday, May 5, 8:30 to 10:20 a.m.
Tuesdays 9:30 a.m.	Wednesday, May 7, 8:30 to 10:20 a.m.
Tuesdays 11 a.m.	Monday, May 5, 11 a.m. to 12:50 p.m.
Tuesdays 12:30 p.m.	Wednesday, May 7, 11 a.m. to 12:50 p.m.
Tuesdays 2 p.m.	Monday, May 5, 1:30 to 3:20 p.m.
Tuesdays 3:30 p.m.	Monday, May 5, 4 to 5:50 p.m.
Tuesdays 5 p.m.	Wednesday, May 7, 4 to 5:50 p.m.
Tuesdays 6:30 p.m.	Monday, May 5, 6:30 to 8:20 p.m.
Tuesdays 8 p.m.	Wednesday, May 7, 6:30 to 8:20 p.m.
**Makeup exams	Friday, May 9, 8:30 a.m. to 12:50 p.m.

*For courses meeting only on Wednesdays or Fridays, follow the Monday schedule. For courses meeting only on Thursdays, follow the Tuesday schedule.

**Makeup exam slot is for students with three or more exams on the same day. These students may postpone one of their exams and take it during this makeup slot. All makeup exams will be held together in the same room and proctored by University personnel.